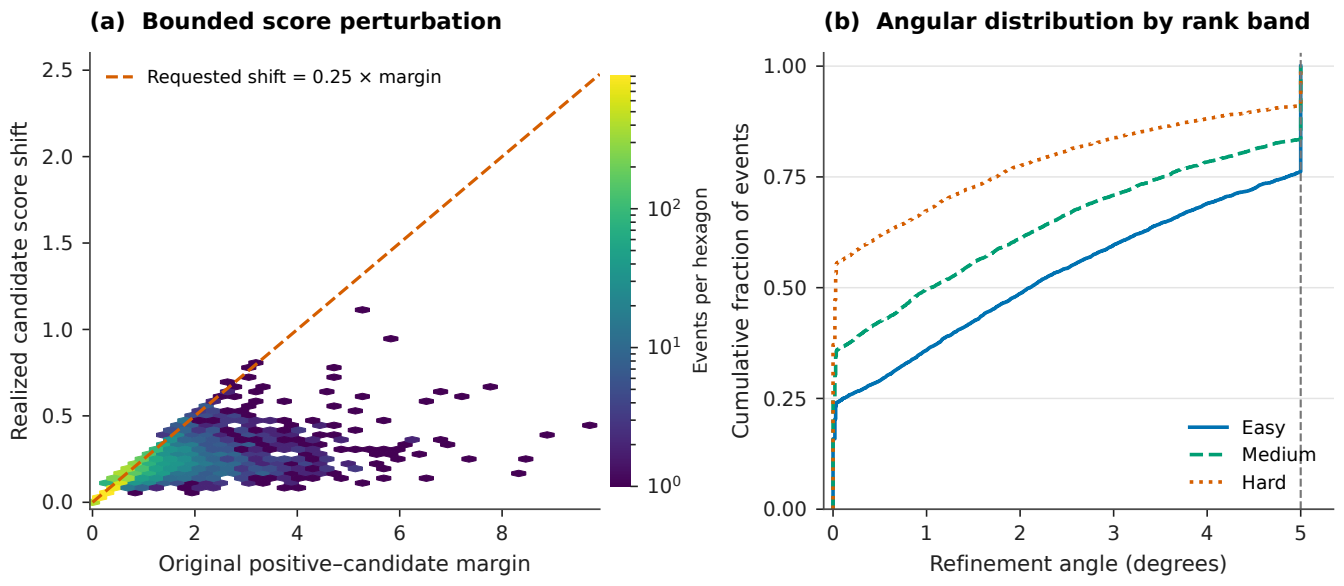

(a) Bounded score perturbation
Requested shift = 0.25 × margin
Realized candidate score shift
Original positive–candidate margin
Events per hexagon
10^0
10^1
10^2
0.0
0.5
1.0
1.5
2.0
2.5
0
2
4
6
8
(b) Angular distribution by rank band
Cumulative fraction of events
Refinement angle (degrees)
Easy
Medium
Hard
0.00
0.25
0.50
0.75
1.00
0
1
2
3
4
5